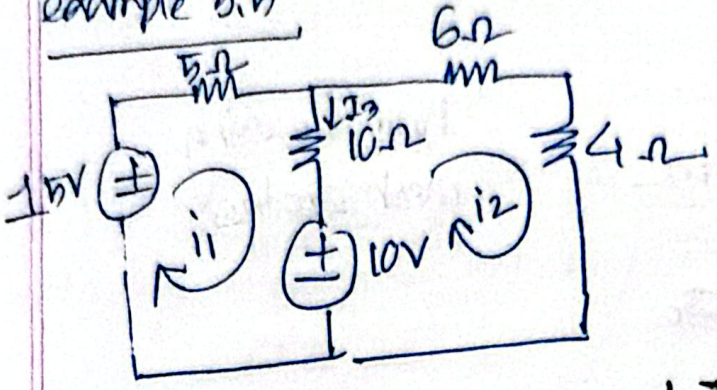
Mesh Analysis

example 3.5



Find I_1, I_2, I_3 using mesh analysis.

In mesh 1,

$-15 + 5i_1 + 10(i_1 - i_2) + 10 = 0$

$\Rightarrow 15i_1 - 10i_2 = 5$

$\Rightarrow \boxed{3i_1 - 2i_2 = 1}$

In mesh 2,

$-10 + 10(i_2 - i_1) + 6i_2 + 4i_2 = 0$

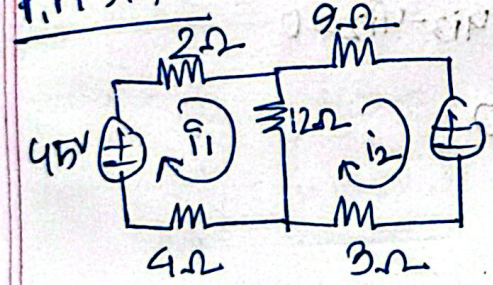
$\Rightarrow -10i_1 + 20i_2 = 10$

$\Rightarrow \boxed{-i_1 + 2i_2 = 1}$

Solving, $i_1 = 1A$, $i_2 = 1A$

$I_3 = i_1 - i_2 = 1 - 1 = 0A$.

P.P. 3.5



Find i_1, i_2 using mesh analysis.

For mesh 1,

$-45 + 2i_1 + 12(i_1 - i_2) + 4i_1 = 0$

$\Rightarrow 18i_1 - 12i_2 = 45$

$\Rightarrow \boxed{6i_1 - 4i_2 = 15}$

In mesh 2,

$9i_2 + 30 + 3i_2 + 12(i_2 - i_1) = 0$

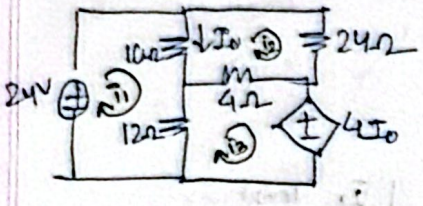
$\Rightarrow 24i_2 - 12i_1 = -30$

$\Rightarrow \boxed{-2i_1 + 4i_2 = -5}$

Solving, $i_1 = 2.5A$

$i_2 = 0A$

Example 3.6



Find I_o using mesh analysis.

$I_o = i_1 - i_2$, $4I_o = 4i_1 - 4i_2$

For mesh 1,

$-24 + 10(i_1 - i_2) + 12(i_1 - i_3) = 0$

$\Rightarrow 6 \cdot 2i_1 - 10i_2 - 12i_3 = 24$

$\Rightarrow \boxed{11i_1 - 5i_2 - 6i_3 = 12}$

For mesh 2,

$24i_2 + 4(i_2 - i_3) + 10(i_2 - i_1) = 0$

$\Rightarrow -10i_1 + 38i_2 - 4i_3 = 0$

$\Rightarrow \boxed{-5i_1 + 19i_2 - 2i_3 = 0}$

For mesh 3,

$4I_o + 12(i_3 - i_1) + 4(i_3 - i_2) = 0$

$\Rightarrow 4i_1 - 4i_2 + 12i_3 - 12i_1 + 4i_3 - 4i_2 = 0$

$\Rightarrow -8i_1 - 8i_2 + 16i_3 = 0$

$\Rightarrow \boxed{-i_1 - i_2 + 2i_3 = 0}$

Solving, $i_1 = 2.25\,A$

$i_2 = 0.75\,A$

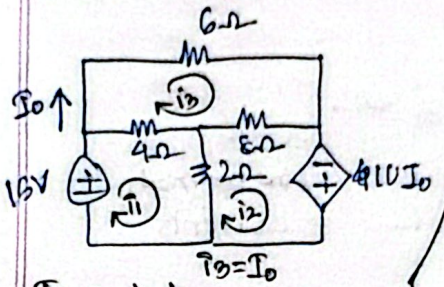
$i_3 = 1.5\,A$

and, $I_o = i_1 - i_2$

$= 2.25 - 0.75$

$= 1.5\,A$

P.P. 3.6



Find I_o using mesh analysis

$i_3 = I_o$

In mesh 1,

$-16 + 4(i_1 - i_3) + 2(i_1 - i_2) = 0$

$\Rightarrow 6i_1 - 2i_2 - 4i_3 = 16$

$\Rightarrow \boxed{3i_1 - i_2 - 2i_3 = 8}$

In mesh 2,

$2(i_2 - i_1) + 8(i_2 - i_3) - 10I_o = 0$

$\Rightarrow -2i_1 + 10i_2 - 8i_3 - 10i_3 = 0$

$\Rightarrow -2i_1 + 10i_2 - 18i_3 = 0$

$\Rightarrow \boxed{-i_1 + 5i_2 - 9i_3 = 0}$

In mesh 3,

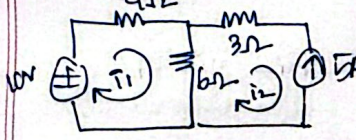
$6i_3 + 8(i_3 - i_2) + 4(i_3 - i_1) = 0$

$\Rightarrow -4i_1 - 8i_2 + 18i_3 = 0$

$\Rightarrow \boxed{-2i_1 - 4i_2 + 9i_3 = 0}$

Solving, $i_1 = \frac{18}{7}\,A$, $i_2 = \frac{54}{7}\,A$, $i_3 = 4\,A$

mesh analysis with current source



Find mesh currents.

$i_2 = -5\,A$

For mesh 1,

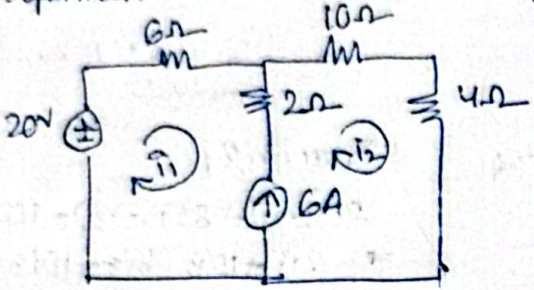
$-10 + 4i_1 + 6(i_1 - i_2) = 0$

$\Rightarrow 10i_1 - 6i_2 = 10$

$\Rightarrow 5i_1 - 3i_2 = 5$

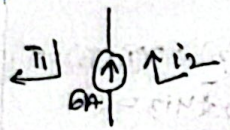
$\Rightarrow i_1 = \frac{5 + 3(-5)}{5} = \frac{5 - 15}{5} = \frac{-10}{5} = -2\,A$

Supermesh: current source shared by 2 meshes.



Find mesh currents.

$$i_2 - i_1 = 6$$

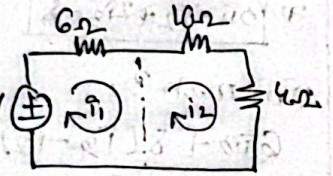


and for supermesh,

$$-20V + 6i_1 + 10i_2 + 4i_2 = 0$$

$$\Rightarrow 6i_1 + 14i_2 = 20$$

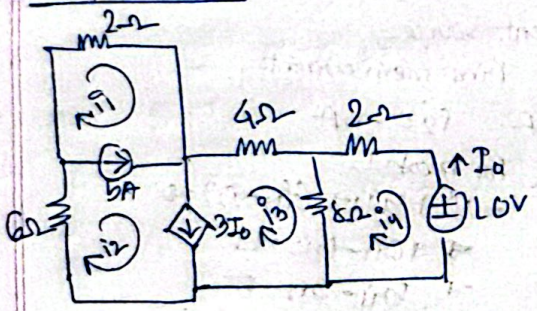
$$\therefore \boxed{3i_1 + 7i_2 = 10}$$



solving,

$i_1 = -3.2A$, $i_2 = 2.8A$

Example 3.7

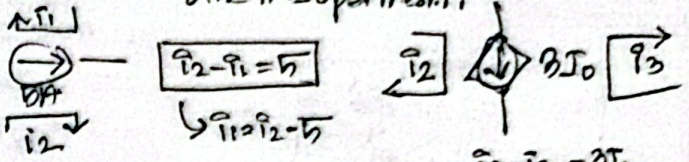


Find i_1 to i_4.

$i_4 = -I_o$

$3I_o = -3i_4$

For mesh 1, 2, 3 are in supermesh.



$$i_2 - i_3 = 3I_o$$

$$\Rightarrow i_2 - i_3 = -3i_4$$

$$\Rightarrow \boxed{i_2 - i_3 + 3i_4 = 0}$$

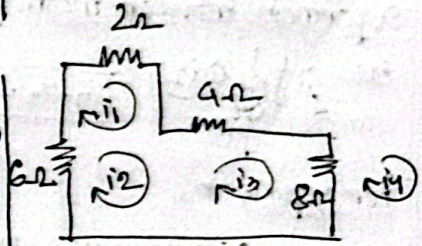
For mesh 4,

$$2i_4 + 10 + 8(i_4 - i_3) = 0$$

$$\Rightarrow -8i_3 + 10i_4 = -10$$

$$\Rightarrow \boxed{-4i_3 + 5i_4 = -5}$$

For supermesh,



$$2i_1 + 4i_3 + 8(i_3 - i_4) + 6i_2 = 0$$

$$\Rightarrow 2i_1 + 6i_2 + 12i_3 - 8i_4 = 0$$

$$\Rightarrow i_1 + 3i_2 + 6i_3 - 4i_4 = 0$$

$$\Rightarrow i_2 - 5 + 3i_2 + 6i_3 - 4i_4 = 0$$

$$\Rightarrow \boxed{4i_2 + 6i_3 - 4i_4 = 5}$$

So solving,

i_2	i_3	i_4	Cons
1	−1	3	0
~~10~~	−4	65	−5
4	6	−4	5

$i_2 = 2.57A$

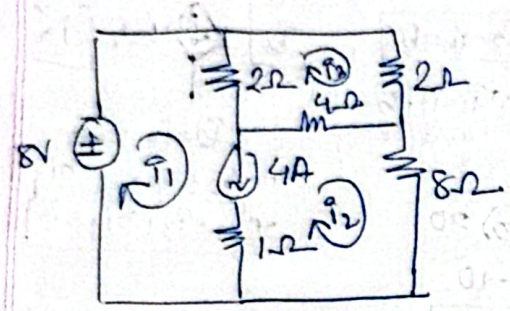
$i_3 = 3.93A$

$i_4 = 2.14$

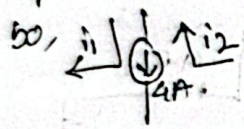
$i_1 = i_2 - 5 = -2.5 - 5 = -7.5A$

P.P. – 3.7

Use mesh analysis to determine i_1, i_2 and i_3.



supermesh between mesh 1 and 2:



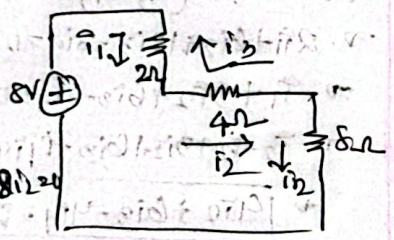
so, $i_1 - i_2 = 4$

for supermesh,

$-8 + 2(i_1 - i_3) + 4(i_2 - i_3) + 8i_2 = 0$

$\Rightarrow 2i_1 + 6i_2 - 6i_3 = 8$

$\Rightarrow \boxed{i_1 + 3i_2 - 3i_3 = 4}$



for mesh 3,

$2(i_3 - i_1) + 2i_3 + 4(i_3 - i_2) = 0$

$\Rightarrow -2i_1 - 4i_2 + 8i_3 = 0$

$\Rightarrow \boxed{-i_1 - 2i_2 + 4i_3 = 0}$

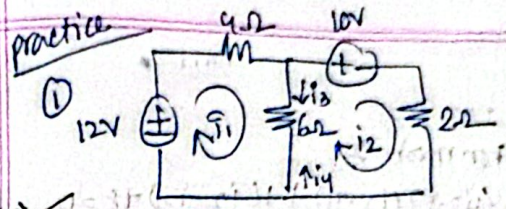
solving equations,

$i_1 = 4.63A$

$i_2 = 0.6315 A = 631.57 mA$

$i_3 = 1.47 A$

practice



Find i_1, i_2, i_3, i_4 using mesh analysis.

Hint:

In mesh 1,

$-12 + 4i_1 + 6(i_1 - i_2) = 0$

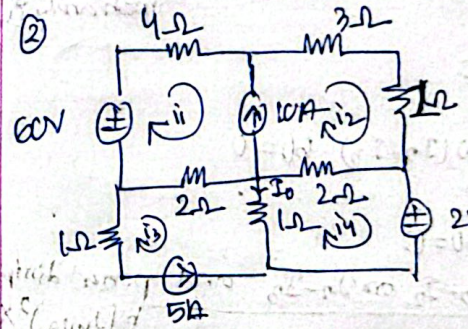
and, $i_3 = i_1 - i_2$

$i_4 = i_2 - i_1$

In mesh 2,

$6(i_2 - i_1) + 10 + 2i_2 = 0$

②



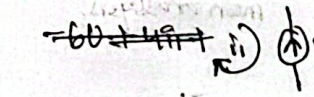
Hint

$I_0 = I_3 - I_4$

In mesh 3, there is current source on the edge.

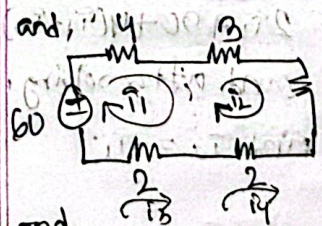
so, $I_3 = -5A$.

for mesh 1, 2 → supermesh,



$i_2 - i_1 = 10$

and,

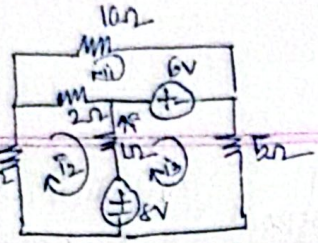


and, $-60 + 4i_1 + 6i_2 + 4i_2 + 2(i_2 - i_4) + 2(i_1 - i_3) = 0$

for mesh 4,

$2(i_4 - i_3) + 2(i_4 - i_2) + 22.5 = 0$

solve proper equations. As there are 4 equations, place value, $i_2 = 10 + i_1$ in each case and get 3 variable 3 equations, then solve them to get i_2, i_3, i_4. Then find i_1 from i_2.



③ Find i_o

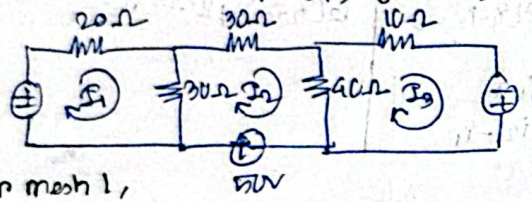
hint: $i_o = i_3 - i_2$

For mesh 1,
$10i_1 - 6 + 2i_1 - i_2 = 0$

For mesh 2,
$4i_2 + 2(i_2 - i_1) + 1(i_2 - i_3) + 8 = 0$

For mesh 3,
$5i_3 - 8 + 1(i_3 - i_2) + 6 = 0$

④



Find power dissipated at 40Ω using mesh analysis.

hint

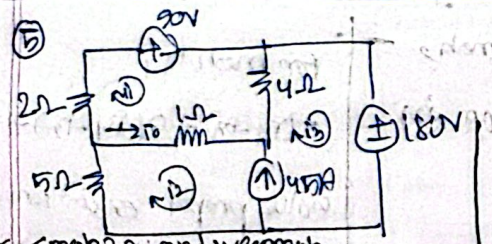
For mesh 1,
$-40 + 20I_1 + 30(I_1 - I_2) = 0$

For mesh 2,
$30(I_2 - I_1) + 30I_2 + 40(I_2 - I_3) - 50 = 0$

For mesh 3,
$4(I_3 - I_2) + 10I_3 - 60 = 0$

solve and get $I_{40\Omega}$ = $I_3 - I_2$ or $I_2 - I_3$ and power dissipated $= (I_{40\Omega})^2 \times R \quad (R = 40)$

⑤



Find i_o using mesh analysis.

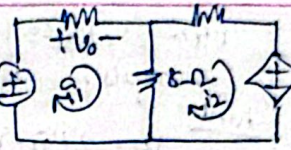
hint: mesh 2,3 and supermesh.
$i_3 - i_2 = 4$

and,
$5i_2 + 1(i_2 - i_1) + 4(i_3 - i_1) + 18 = 0$

For mesh 1,
$2i_1 + 30 + 4(i_1 - i_3) = 0$

and after solving,
find $i_o = -i_1$

⑥



calculate the mesh currents.

hint: Mesh currents are i_1 and i_2.

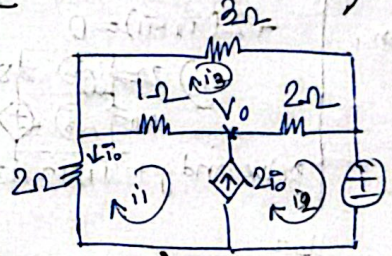
$V_o = 3i_1$, $2V_o = 2 \times 3i_1 = 6i_1$

For mesh 1,
$-12 + 3i_1 + 6(i_1 - i_2) = 0$

For mesh 2,
$6(i_2 - i_1) + 6i_2 + 2V_o = 0$
$\Rightarrow 6(i_2 - i_1) + 6i_2 + 6i_1 = 0$

(solve and get i_1 and i_2)

⑦



Find V_o and i_o using mesh analysis.

solution hint: $i_o = -i_1$, $2i_o = -2i_1$

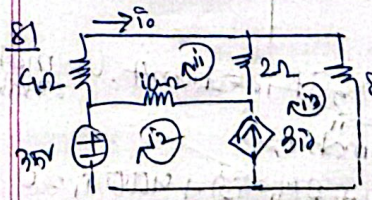
In mesh 1, 2 → supermesh
$2i_1 + 1(i_1 - i_3) + 2(i_2 - i_3) + 27 = 0$
$\Rightarrow 3i_1 + i_2 + 2i_3 = -27$

and, $2i_o = i_2 - i_1$ $\quad -i_3$
$\Rightarrow -2i_1 = i_2 - i_1$
$\Rightarrow 0 = i_2 + i_1$
$\therefore i_2 + i_2 = 0$ $\quad i_1 + i_2 = 0$

For mesh 3,
$3i_3 + 2(i_3 - i_2) + 1(i_3 - i_1) = 0$
$\Rightarrow -i_1 + 2i_2 + 6i_3 = 0$

solving, $i_1 = -18$ A, $i_2 = 18$ A, $i_3 = 3$ A
$\therefore i_o = -i_1 = 2 \cancel{0A} - (18) = 18$ A
$\therefore V_o = 1(i_3 - i_1) = 3 + 18 = 21$ V

⑧



Find i_o using mesh analysis.

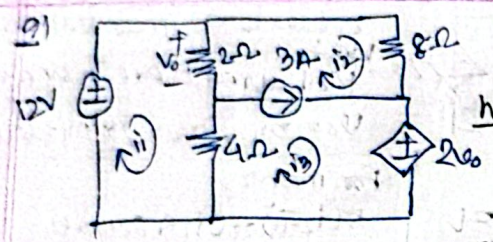
hint: $i_3 - i_2 = 8i_o$, $i_o = i_1$
$\Rightarrow i_3 - i_2 = 8i_1$
$\Rightarrow 8i_1 + i_2 - i_3 = 0$

Supermesh between mesh 2,3,
$-35 + 10(i_2 - i_1) + 2(i_3 - i_1) + 8i_3 = 0$

For mesh 1,
$4i_1 + 2(i_1 - i_3) + 10(i_1 - i_2) = 0$

solve and get i_1, i_2, i_3 and $i_o = i_1$.

9)



Find i_1, i_2, i_3 using mesh analysis.

hint

$V_0 = 2(i_1 - i_2)$

$\therefore 2V_0 = 4i_1 - 4i_2$

In mesh 1,

$-12 + 2(i_1 - i_2) + 4(i_1 - i_3) = 0$

In mesh 2, 3 → supermesh.

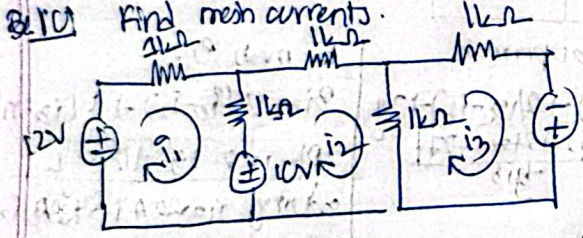
~~$i_2 - i_3 = 3$~~ $i_3 - i_2 = 3$

Also for supermesh,

$8i_2 + (2V_0 \to 4i_1 - 4i_2) + 4(i_3 - i_1) + 2(i_2 - i_1) = 0$

solve and get i_1, i_2, i_3.

10) Find mesh currents.



hint

In mesh 1,

$-12 + 1000(i_1 - i_2) + 1000i_1 + 10 = 0$

In mesh 2,

$-10 + 1000(i_2 - i_1) + 1000i_2 + 1000(i_2 - i_3) = 0$

In mesh 3,

$1000(i_3 - i_2) + 1000i_3 - 12 = 0$

11) Find value of R, V_1, V_2 if $i_0 = 15mA$. Use mesh analysis



$i_0 = i_1$

for mesh 1,

$-90 + R(i_1 - i_2) + 4000i_1 = 0$

$\Rightarrow Ri_1 - Ri_2 + (R + 4000)i_1 = 90$

$i_2 = i_0 = 15mA$.

So, $(R + 4000) \times 15 \times 10^{-3} - Ri_2 = 90$

$\Rightarrow 15R \times 10^{-3} + 60 - Ri_2 = 90$

$\Rightarrow R(15 \times 10^{-3} - i_2) = 90 - 60 = 30$

$\therefore R = \frac{30}{15 \times 10^{-3} - i_2}$

For mesh 2,

$R(i_2 - i_1) + 3000i_2 = 0$

$\Rightarrow R = \frac{-3000i_2}{i_2 - i_1} = \frac{-3000i_2}{i_2 - 15 \times 10^{-3}}$

Now,

$\frac{30}{15 \times 10^{-3} - i_2} = \frac{-3000i_2}{i_2 - 15 \times 10^{-3}}$

$\Rightarrow \frac{2(i_2 - 15 \times 10^{-3})}{(15 \times 10^{-3} - i_2)} = -\frac{3000i_2}{2}$

$\Rightarrow \frac{-(15 \times 10^{-3} - i_2)}{(15 \times 10^{-3} - i_2)} = -1000i_2$

$\Rightarrow -1 = -1000i_2$

$\therefore i_2 = \frac{1}{1000} = 0.01A$

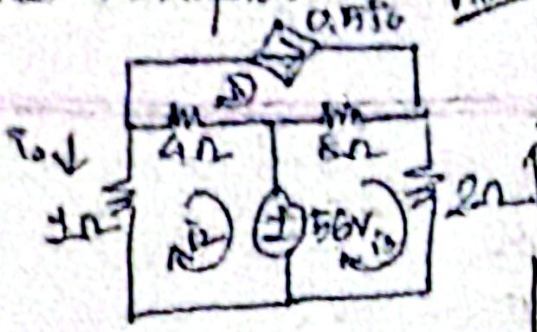
$\therefore R = \frac{30}{15 \times 10^{-3} - 0.01} = 6000\Omega = 6k\Omega$

$\therefore V_1 = i_2 \times 3000 = 0.01 \times 3000 = 30V$.

$\therefore V_2 = i_1 \times 4000 = 15 \times 10^{-3} \times 3000 = 45V$.

Scanned with CamScanner

[11] Calculate the power dissipated in each resistor use mesh analysis.



Hint

$i_2 = -i_0 = 0.5i_0 = -0.5i_2$

In mesh 1,

$4(i_2 - i_1) + 56 = 0$

In mesh 2,

$i_1 = 0.5i_0 = -0.5i_2$

$\therefore i_1 + 0.5i_2 = 0$

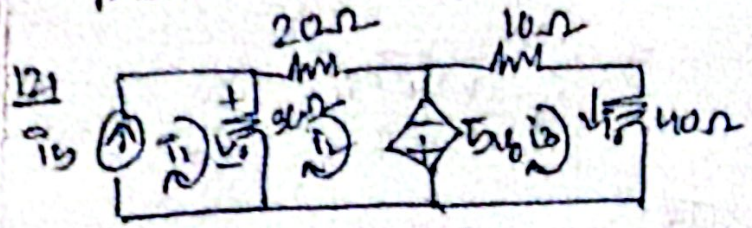
In mesh 3,

$-56 + 8(i_3 - i_1) + 2i_3 = 0$

solve eqn and get i_1, i_2, i_3.

Find power dissipated, $P_{4\Omega} = (i_2 - i_1)^2 \times 4$, $P_{8\Omega} = (i_3 - i_1)^2 8$, $P_{2\Omega} = i_3^2 \times 2$

[12]



Find i_o/i_s value using mesh analysis.

$i_o = i_3$

$V_0 = (i_1 \times 30)$

$= 30i_1 - 30i_2$

$5V_0 = 5 \times 6(i_1 + i_2)$

$= 150i_1 + 150i_2$

In mesh 1,

$i_1 = i_s$

In mesh 2,

$30(i_2 - i_1) + 20i_2 - 5V_0 = 0$

$\Rightarrow -30i_1 + 50i_2 - 150i_1 + 150i_2 = 0$

$\Rightarrow \boxed{-180i_1 + 200i_2 = 0}$

In mesh 3,

$5V_0 + 10i_3 + 40i_3 = 0$

$\Rightarrow \boxed{150i_1 + 150i_2 + 50i_3 = 0}$

We need $i_o/i_s = i_3/i_1$

So,

$-180i_1 + 200i_2 = 0$

$\Rightarrow -180i_1 = -200i_2$

$\Rightarrow i_2 = \frac{+180}{+200} i_1$

$= \frac{+9}{10} i_1$

$\Rightarrow i_2 = 0.9i_1$

$150i_1 - 150i_2 + 50i_3 = 0$

$\Rightarrow 150i_1 - 150(\frac{+9}{10})i_1 + 50i_3 = 0$

$\Rightarrow 150i_1 - 135i_1 + 50i_3 = 0$

$\Rightarrow 150i_1 + 50i_3 = 135$

$\Rightarrow 30i_1 + 10i_3 = 27$

$150i_1 + 50i_3 = 0$

$\Rightarrow 150i_1 = -50i_3$

$\Rightarrow \frac{i_3}{i_1} = -\frac{150}{50}$

$= \frac{3}{10}$

$= 0.3$